

Markscheme

Mathematics: analysis and approaches
Practice question bank

75. (a)

A theatre has 25 seats in the first row, and each subsequent row has 3 more seats than the previous row. The theatre has 30 rows.

Find the total number of seats in the theatre.

identify an arithmetic model: $u_1 = 25$, $d = 3$, $n = 30$ *(M1)*

$$S_{30} = \frac{30}{2} (2(25) + 29(3)) \quad \text{*(M1)*}$$

$$= 2055 \quad \text{A1}$$

[4 marks]